

Thermodynamic State Vector Interface and Entropy Generation Auditing in the Web of Life Simulation Engine

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Sprint 010 Technical Report

Abstract

We present the architectural formalization and implementation of thermodynamic state vectors, entropy generation auditing, and exergy destruction mechanics for the **Web of Life** simulation engine (Sprint 010). Building upon foundational biogeochemical cycles (Carbon, Nitrogen, Phosphorus, and Water) and open-system Earth Pod models, this sprint introduces rigorous type contracts (`src/thermodynamics/types.ts`) and monad-based evaluation pipelines that enforce the First and Second Laws of Thermodynamics at every simulation epoch. By treating the Earth Pod as a non-equilibrium open thermodynamic system driven by solar exergy influx and governed by internal metabolic dissipation, our framework explicitly tracks internal energy stocks, thermal gradients, boundary fluxes, and irreversible entropy production. We derive the governing balance equations—incorporating the Clausius inequality and the Gouy-Stodola theorem—and detail runtime validation guards that intercept thermodynamic violations. This work establishes a formal thermodynamic foundation for computational systems ecology, ensuring that simulated biospheric metabolisms remain physically consistent with non-equilibrium thermodynamics.

Keywords: Non-equilibrium thermodynamics, entropy generation rate, exergy destruction, Earth Pod, Web of Life, systems ecology, thermodynamic state vectors.

1 Introduction and Systems Ecology Framing

In computational ecosystems and biosphere simulations, maintaining physical rigor is paramount. Models that track nutrient cycling (C , N , P , H_2O) without enforcing thermodynamic constraints risk generating unphysical energy dynamics or violating fundamental conservation laws. Within the **Web of Life** framework, the simulated biosphere is modeled as an **Earth Pod**: an open thermodynamic system that exchanges energy (via solar shortwave radiation, thermal longwave emission, sensible heat, and latent evapotranspirative fluxes) with its external environment while conserving total mass.

Sprint 010 establishes strict interface contracts and runtime verification guards to govern these energy transformations. By framing ecological metabolism as an irreversible thermodynamic engine, we quantify how biological work degrades available energy into thermal dissipation, driving internal entropy generation ($\dot{S}_{\text{gen}}$) and exergy destruction ($\dot{I}$).

Official source code, specifications, and test suites are maintained in the canonical repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

2 Thermodynamic Foundations

2.1 First Law: Energy Conservation

For the Earth Pod open thermodynamic system, the time rate of change of total internal energy (U_{sys}) is balanced by boundary radiative, conductive, and convective fluxes:

$$\frac{dU_{\text{sys}}}{dt} = \Phi_{\text{solar}} - \Phi_{\text{thermal}} - \Phi_{\text{sensible}} - \Phi_{\text{latent}} + \sum_k \dot{W}_k \quad (1)$$

Where:

- U_{sys} : Total internal energy of all biogeochemical stocks [J].
- Φ_{solar} : Incoming shortwave solar radiation flux [W].
- Φ_{thermal} : Outgoing longwave thermal radiation flux [W].
- Φ_{sensible} : Boundary sensible heat flux [W].
- Φ_{latent} : Latent heat flux due to phase changes [W].
- $\dot{W}_k$: Net mechanical/biochemical work transfer (assumed zero for closed-mass biomes) [W].

2.2 Second Law: Entropy Balance & Exergy Destruction

The Second Law mandates that the entropy change of the system accounts for boundary entropy exchanges and internal irreversible generation:

$$\frac{dS_{\text{sys}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{\text{gen}} \quad (2)$$

where $\dot{S}_{\text{gen}} \geq 0$ (Clausius inequality). Solving for the internal entropy generation rate:

$$\dot{S}_{\text{gen}} = \frac{dS_{\text{sys}}}{dt} - \left(\frac{\Phi_{\text{solar}}}{T_{\text{sun}}} - \frac{\Phi_{\text{thermal}}}{T_{\text{sys}}} - \frac{\Phi_{\text{sensible}}}{T_0} - \frac{\Phi_{\text{latent}}}{T_{\text{phase}}} \right) \geq 0 \quad (3)$$

Invoking the **Gouy-Stodola theorem**, the exergy destruction rate ($\dot{I}$) quantifies the lost work potential due to thermodynamic irreversibility, scaled by the reference ambient temperature T_0 :

$$\dot{I} = T_0 \cdot \dot{S}_{\text{gen}} \geq 0 \quad [\text{W}] \quad (4)$$

3 Interface Contracts & Executable Architecture

The core data structures are implemented in `src/thermodynamics/types.ts` and evaluated within `src/thermodynamics/thermodynamic_structure.ts`.

3.1 TypeScript Type Definitions

```
export interface BoundaryFluxVector {
  solarRadiationIn: number;    // [W] Incoming solar shortwave flux
  thermalRadiationOut: number; // [W] Outgoing longwave thermal
    radiation
  sensibleHeatFlux: number;    // [W] Convective/conductive boundary
    sensible heat
```

```

    latentHeatFlux: number;      // [W] Evapotranspirative latent heat
    flux
    netMassEnthalpyFlux: number; // [W] Net enthalpy carried by boundary
    mass transfers
}

export interface ThermodynamicStateVector {
    timestamp: number;           // [s] Simulation epoch time
    ambientTemperature: number; // T_0 [K] Reference ambient temperature
    systemTemperature: number;   // T [K] Effective internal system
    temperature
    internalEnergy: number;      // U [J] Total internal energy of system
    stocks
    totalEntropy: number;        // S [J/K] Total system entropy
    entropyGenerationRate: number; // S_gen_dot [W/K] Internal
    irreversible entropy generation rate
    exergyDestructionRate: number; // I_dot [W] Exergy destruction rate (
    T_0 * S_gen_dot)
    boundaryFluxes: BoundaryFluxVector;
}

export interface IThermodynamicSystem {
    getStateVector(): ThermodynamicStateVector;
    computeEntropyGeneration(dt: number): number;
    verifySecondLaw(): boolean;
}

```

3.2 Runtime Verification Guard

To prevent unphysical states during long-term simulation runs, the engine enforces strict runtime assertions:

```

export function assertSecondLaw(state: ThermodynamicStateVector):
    boolean {
    const EPSILON = 1e-9;
    if (state.entropyGenerationRate < -EPSILON) {
        throw new Error('CRITICAL THERMODYNAMIC VIOLATION: S_gen_dot < 0');
    }
    if (state.exergyDestructionRate < -EPSILON) {
        throw new Error('CRITICAL THERMODYNAMIC VIOLATION: I_dot < 0');
    }
    const expectedExergy = state.ambientTemperature * state.
        entropyGenerationRate;
    if (Math.abs(state.exergyDestructionRate - expectedExergy) > 1e-5) {
        throw new Error('EXERGY CONSISTENCY FAILURE: I_dot != T_0 *
            S_gen_dot');
    }
    return true;
}

```

4 Verification and Audit Matrix

5 Conclusion & Future Outlook

Sprint 010 establishes a mathematically rigorous thermodynamic backbone for the **Web of Life** simulation engine. By formally tracking internal energy, entropy, boundary fluxes, and

Test Case	Condition	Expected Outcome	Status
TC-01	High solar radiation, standard metabolism	$\dot{S}_{\text{gen}} > 0, \dot{I} > 0$	PASSED
TC-02	Zero-flux equilibrium thermal state	$\dot{S}_{\text{gen}} \approx 0, \dot{I} \approx 0$	PASSED
TC-03	Spurious negative entropy injection	Throws CRITICAL VIOLATION	PASSED (Caught)

Table 1: Thermodynamic audit and verification matrix implemented in `tests/sprint_010.test.ts`.

exergy destruction, the engine ensures that all simulated ecological dynamics adhere to the fundamental laws of physics. Future sprints will expand upon these foundations by integrating spatial thermodynamic gradients across regional biomes and coupling exergy efficiency metrics directly to evolutionary selection pressures.

For complete source code and commit history, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>